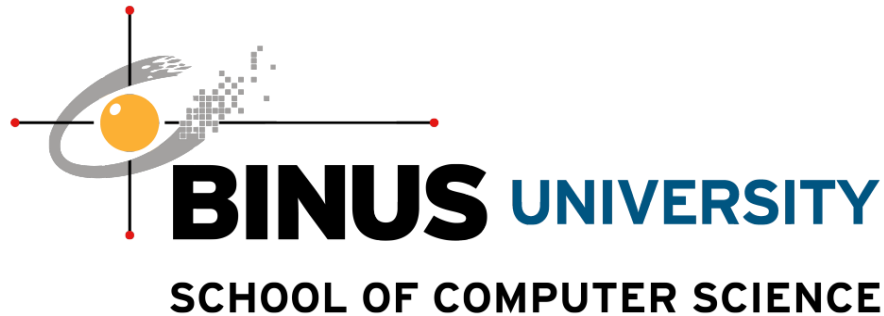


DOCUMENTATION
DATA MINING & VISUALIZATION



Group 3 LC09

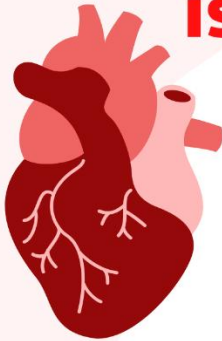
Member :

- | | |
|-------------------------|----------------|
| 1. Beatrice Nathania | Nim 2802490946 |
| 2. Kelly Natalia | Nim 2802472426 |
| 3. Callista Gianna Then | Nim 2802498293 |
| 4. Wilson Iskandar | Nim 2802474343 |
| 5. Felicia Tiffany | Nim 2802456712 |
| 6. Almayra Edgina Rahma | Nim 2802460930 |
| 7. Miryam Almira Levina | Nim 2802465944 |

Dataset : <https://www.kaggle.com/datasets/fedesoriano/heart-failure-prediction>

Bina Nusantara University Jl. Raya Kb. Jeruk No.27, RT.1/RW.9, Kemanggisian, Kec. Palmerah,
Kota Jakarta Barat, Daerah Khusus Ibukota Jakarta 11530

IS CHOLESTEROL THE CULPRIT BEHIND HEART DISEASE?



What is heart disease?

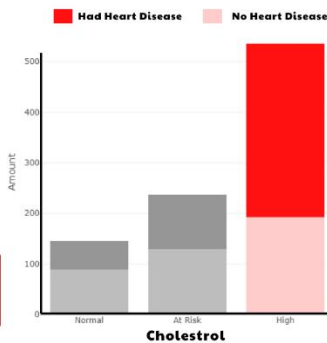
conditions that affect the **heart** and blood vessels, **reducing** the heart's **ability to function properly**

Symptoms

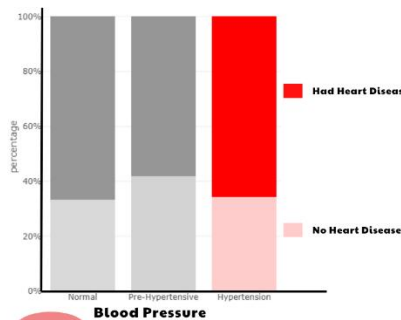


Heart Disease Main Factor

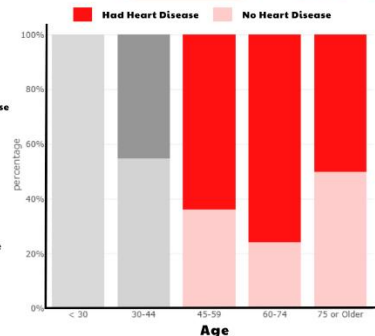
1. High Cholesterol



2. High Blood Pressure



3. Age

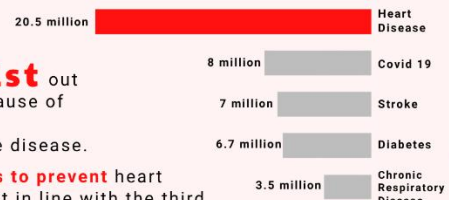


Where the data come from?

The patient data are collected from 4 institutions that specialize in heart disease and health research.

Is heart disease really the biggest threat?

Non-Communicable Disease Death Rate- 2019

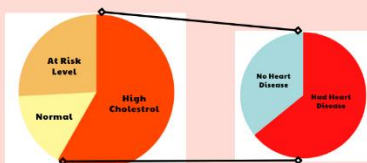


Standing at **1st** out of the top 5 cause of death by non-communicable disease.

The **challenges to prevent** heart disease make it in line with the third SDGs

Cholesterol Caused Heart Disease? Is it true?

Absolutely true! In conclusion, the data shows that patients with high cholesterol levels **are more likely to develop heart disease**.



How to prevent heart disease?



DOES CHOLESTEROL CAUSE HEART DISEASE?

3rd SDGs Good Health and Well-Being

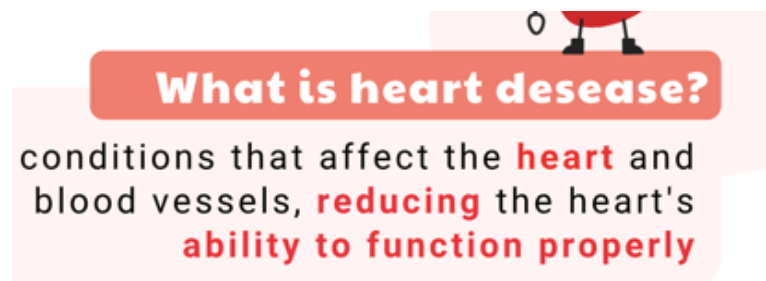


The Sustainable Development Goals (SDGs), established by the United Nations, aim to address global challenges and ensure a better and more sustainable future for all. The third goal in SDG is to ensure healthy lives and promote well-being for all at all ages. It promotes healthy lifestyles, preventive measures, and modern, and efficient healthcare for everyone. Significant strides were made in increasing life expectancy and reducing some of the common killers associated with child and maternal mortality. But more efforts are needed to fully eradicate a wide range of diseases and address many different persistent and emerging health issues. By focusing on providing more efficient funding of health systems, improved sanitation and hygiene, and increased access to physicians, significant progress can be made in helping to save the lives of millions.

Heart disease intersects with the third SDG (Good Health and Well-being) by being the single largest contributor to premature mortality from non-communicable diseases. As the leading cause of death among the top non-communicable diseases with 20.5 million deaths, it places an immense burden on global health systems and impacts health equity and access to healthcare. Therefore, addressing heart disease is crucial for achieving SDG 3's goal of ensuring healthy lives, promoting well-being for all, and reducing premature mortality through prevention, treatment, and the promotion of a healthy lifestyle.

https://sdgs.un.org/goals/goal3#targets_and_indicators

Definition and Type of Diabetes

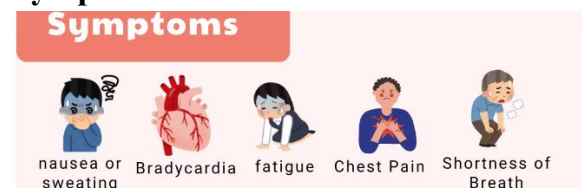


Heart disease refers to a range of conditions that affect the heart and blood vessels, which reduces the heart's ability to function properly. The condition can manifest through several key symptoms, which include nausea or sweating, palpitations, fatigue, chest pain, and shortness of breath. While there are various risk factors, the poster highlights three main contributing factors associated with the development of heart disease. These include high cholesterol, which shows a strong correlation with the presence of heart disease; high blood pressure, where a majority of patients with hypertension also have the disease; and advanced age, as the prevalence of heart disease increases significantly in older populations.

<https://www.mayoclinic.org/diseases-conditions/heart-disease/symptoms-causes/syc-20353118>

<https://www.cdc.gov/heart-disease/risk-factors/index.html#:~:text=Overview,Smoking1>

Symptoms of Heart Disease

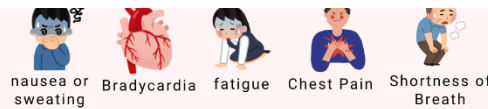


It is important to recognize the key symptoms that can act as warning signs for heart disease. The specific physical indicators highlighted are nausea or sweating, palpitations, fatigue, chest pain, and shortness of breath.

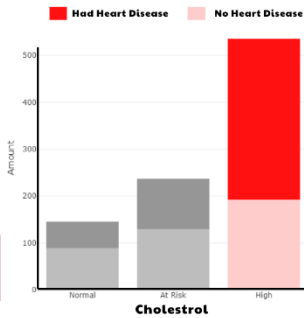
<https://www.mayoclinic.org/diseases-conditions/heart-disease/symptoms-causes/syc-20353118>

Heart Disease Main Factors

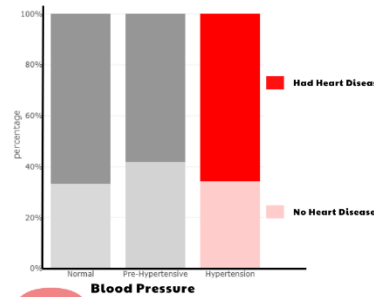
Heart Disease Main Factor



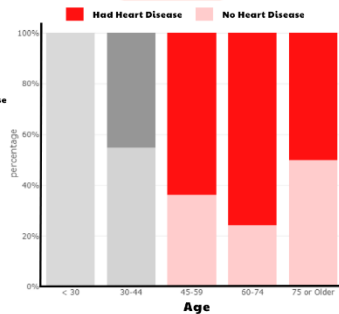
1. High Cholesterol



2. High Blood Pressure



3. Age

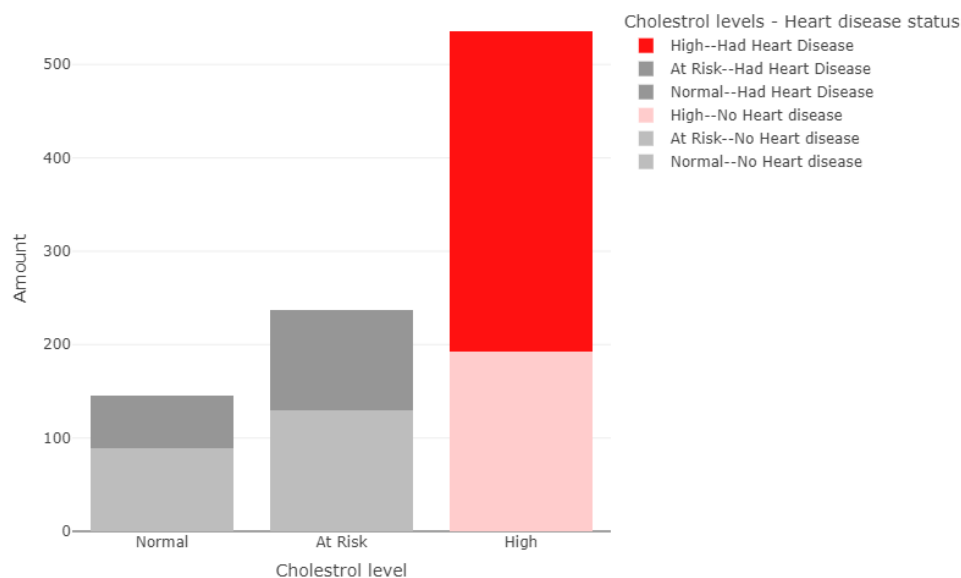


The dataset reveals three main factors that show a strong correlation with the presence of heart disease. These are High Cholesterol, High Blood Pressure, and Age.

1. High Cholesterol

The bar chart for cholesterol indicates a clear relationship between cholesterol levels and heart disease. In the category for "High" cholesterol, the number of patients with heart disease is substantially greater than those without it. This suggests, based on the data analyzed, that high cholesterol is a significant contributing factor to developing heart disease.

Patient Heart disease status divided by cholesterol levels



Here is a stacked bar chart that shows the distribution of cholesterol level from patients with heart disease.

- Level Categories :

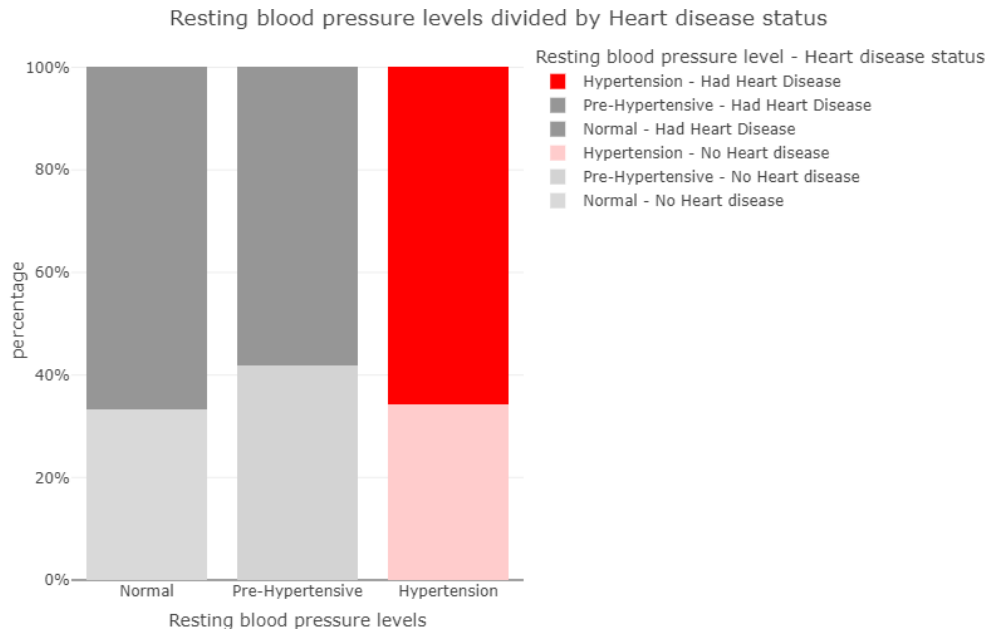
- Normal
Serum cholesterol below 200 mg/dl
 - At risk
Serum cholesterol between 200 mg/dl to 239 mg/dl
 - High
Serum cholesterol more than 240 mg/dl
- Using barchart allowed easier overview of the data, in this barchart :
 - Each categories in x-axis represent Cholesterol Level
 - With a y-axis that represent the amount of patient that suffered from cholesterol
 - This barchart shows the number of patients with a certain heart status for each level of cholesterol.
 - Normal Level
 - Had Heart Disease : 57
 - No Heart disease : 89
 - At risk Level
 - Had Heart Disease : 108
 - No Heart disease : 129
 - High Level
 - Had Heart Disease : 343
 - No Heart Disease : 192
 - Insight :

From the data, most patients with high cholesterol have heart disease. This indicates that cholesterol is an important factor that causes heart disease. It increases the risk of getting a heart disease significantly. The ratio of no heart disease to heart disease per category of cholesterol levels are:

 - Normal: 1 : 1.561
 - At Risk: 1 : 1.194
 - High : 1: 1.786

2. High Blood Pressure

The data that was filtered for people with a high cholesterol level, shows that a very high percentage of individuals with hypertension also have heart disease. For the hypertension group, the proportion of people with heart disease is significantly larger than those without, highlighting high blood pressure as a major factor.



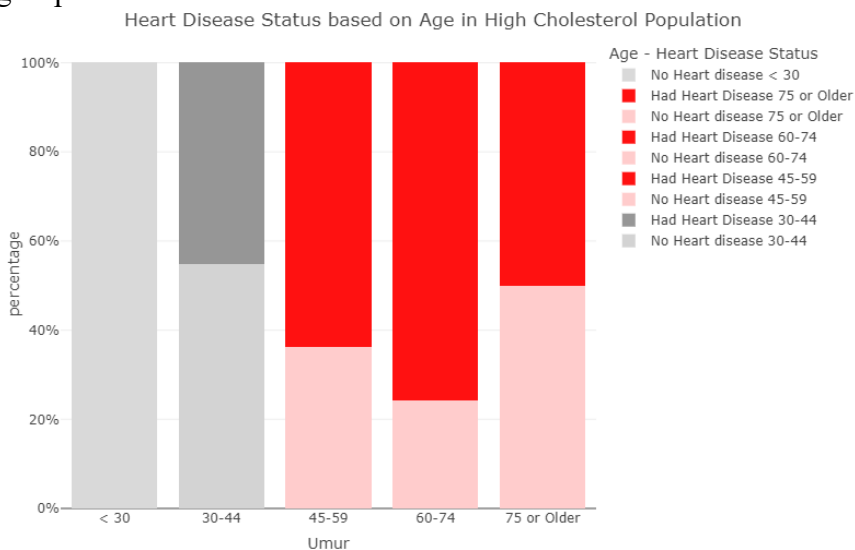
- **Level Categories :**
Level categories were taken from a cleaned variable, RestingBP_T from
 - Normal
RestingBP_T is less than 120 mm Hg
 - Pre-Hypertensive
RestingBP_T is in between 120 mm Hg and 130 mm Hg
 - Hypertension
RestingBP_T is 130 mm Hg above
- Using a stacked barchart allowed easier overview of the data, in this barchart :
 - Each categories in x-axis represent Blood Pressure Level
 - With the y-axis representing the percentage of patients that suffered from heart diseases.
- This barchart shows the percentage of patients with high cholesterol's heart disease status for each level of blood pressure, rounded to three significant figures.
 - Normal Level
 - Had Heart Disease : 66.7%
 - No Heart disease : 33.3%
 - Pre-Hypertensive
 - Had Heart Disease : 58.1%
 - No Heart disease : 41.9%
 - Hypertension
 - Had Heart Disease : 65.7%
 - No Heart Disease : 34.3%

- Insight :

According to the results, the majority of the patients have heart disease for all categories of blood pressure, but looking into the details, the trend fluctuates. The blood pressure status with the largest percentage of having a heart disease is normal level. Followed by hypertension patients and finally with the smallest percentage, pre-hypertensive levels. This fluctuation tells us that cholesterol is still a more significant impact than blood pressure on heart disease.

3. Age

A clear trend is visible in the analysis of age groups, indicating that the risk of heart disease increases with age. The charts show that for older age brackets, specifically those 55 years and older, the percentage of individuals with heart disease is considerably higher than in the younger groups.



- Age Categories :

Age categories were taken from variable Age from the dataset

- < 30
Age was recorded to be less than 30 “< 30”
- 30-44
Age was recorded be between 30 and 45 (Age $\geq$ 30 and Age < 45)
- 60-74
Age was recorded between 60 and 75 (Age $\geq$ 60 and Age < 75)
- 75 or Older
Age was recorded to be more than 75 (Age $\geq$ 75)

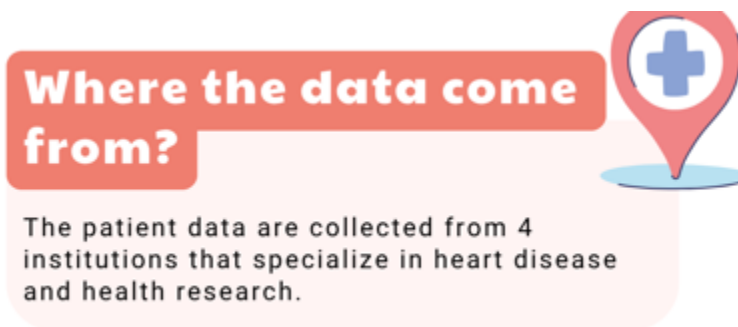
- Using a stacked barchart allowed easier overview of the data, in this barchart :

- Each categories in x-axis represent Age

- With the y-axis representing the percentage of patients that suffered from heart diseases
- This barchart shows the percentage of patients with high cholesterol's heart disease status for each age category, rounded to three significant figures
 - - Had Heart Disease : 0.00%
 - No Heart disease : 100%
 - 30-44
 - Had Heart Disease : 45.1%
 - No Heart disease : 54.9%
 - 45-59
 - Had Heart Disease : 63.7%
 - No Heart Disease : 36.3%
 - 60-74
 - Had Heart Disease : 75.9%
 - No Heart disease : 24.1%
 - 75 or Older
 - Had Heart Disease : 50.0%
 - No Heart disease : 50.0%
- Insight :

Based on the results, people aged under 30, do not have heart disease and there is a clear trend that heart disease percentage increases with age, however the trend drops at age 75 or older, this may be caused because of the high mortality rate of seniors with heart diseases which contribute to a lower overall percentage. Age is a significant factor of heart disease.

Data Source

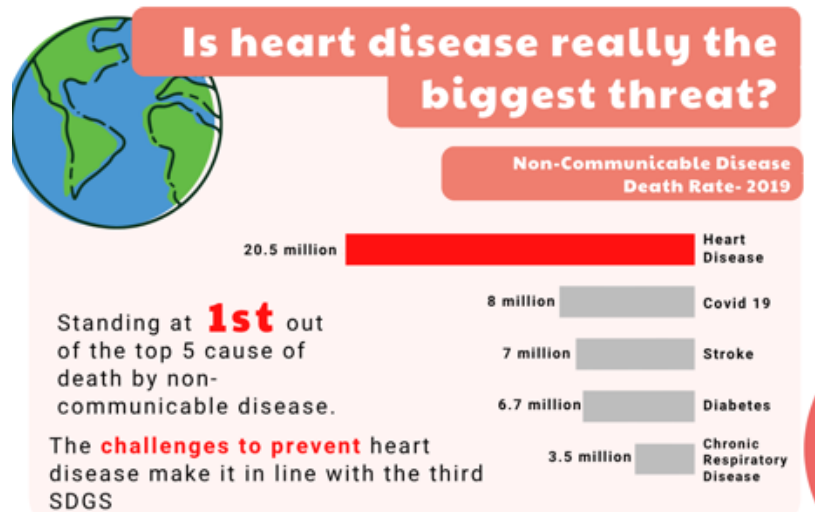


To ensure the analysis is based on credible and relevant information, the poster clarifies where the data for the study was obtained. The patient data used in this analysis was collected from

four institutions that specialize in heart disease and health research. This specialized sourcing suggests that the dataset is focused and relevant to the study of heart conditions.

<https://www.kaggle.com/datasets/fedesoriano/heart-failure-prediction>

The Global Impact of Heart Disease



Our research addresses the critical question of whether heart disease is the biggest global health threat, and the findings confirm that it is. An analysis of 2019 death rates for non-communicable diseases reveals that heart disease is the leading cause of death among the top five conditions, ranking at number one.

It was responsible for 20.5 million deaths, a figure that significantly surpasses other major causes. For comparison:

- Covid-19 accounted for 8 million deaths.
- Stroke accounted for 7 million deaths.
- Diabetes accounted for 6.7 million deaths.
- Chronic Respiratory Disease accounted for 3.5 million deaths.

Given the scale of this public health issue, the challenges involved in preventing heart disease are directly in line with the mission of the 3rd Sustainable Development Goal (SDG) for Good Health and Well-being.

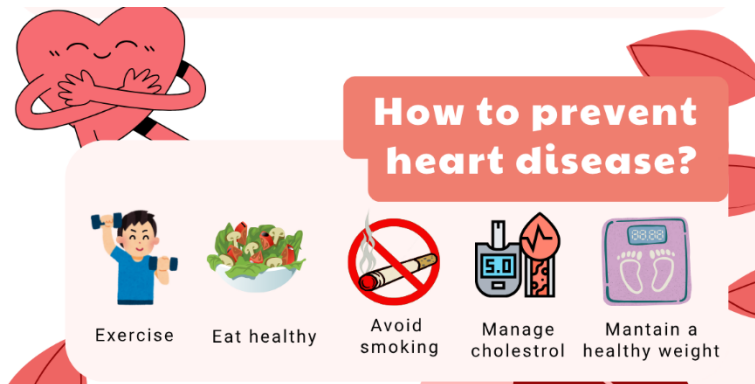
<https://www.who.int/news-room/fact-sheets/detail/the-top-10-causes-of-death>

<https://www.ncbi.nlm.nih.gov/books/NBK581940/#:~:text=Diabetes%20is%20a%20major%20d river,deaths%20in%20people%20under%2060.>

[https://www.who.int/news-room/fact-sheets/detail/chronic-obstructive-pulmonary-disease-\(copd\)](https://www.who.int/news-room/fact-sheets/detail/chronic-obstructive-pulmonary-disease-(copd))

[https://pmc.ncbi.nlm.nih.gov/articles/PMC10809869/#:~:text=Introduction-Cardiovascular%20diseases%20\(CVDs\)%20are%20the%20leading%20cause%20of%20mortality%20globally,progressing%20more%20slowly%20in%20LMICs.](https://pmc.ncbi.nlm.nih.gov/articles/PMC10809869/#:~:text=Introduction-Cardiovascular%20diseases%20(CVDs)%20are%20the%20leading%20cause%20of%20mortality%20globally,progressing%20more%20slowly%20in%20LMICs.)

Recommendations for Heart Disease Prevention



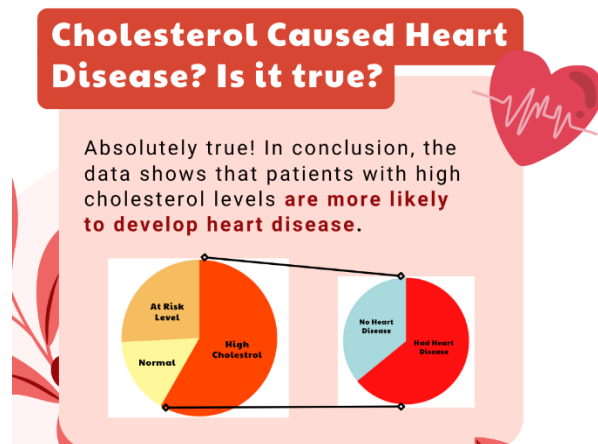
To actively reduce the risk of developing heart disease, our findings point to five key preventative measures. These lifestyle and health management strategies are crucial for maintaining cardiovascular well-being.

The recommendations are as follows:

- Exercise: Engage in regular physical activity.
- Eat healthy: Adopt and maintain a healthy diet.
- Avoid smoking: It is critical to avoid smoking.
- Manage cholesterol: Actively monitor and manage cholesterol levels.
- Maintain a healthy weight: Ensure you maintain a body weight that is within a healthy range.

<https://www.mayoclinic.org/diseases-conditions/heart-disease/in-depth/heart-disease-prevention/art-20046502>

Conclusion: The Link Between Cholesterol and Heart Disease



Our research aimed to answer the critical question: "Does cholesterol cause heart disease?" The conclusion drawn from our data analysis is definitive: it is "Absolutely true".

The findings show a clear and strong correlation, leading to the conclusion that patients with high cholesterol levels are more likely to develop heart disease. This is visually supported by our data breakdown. The analysis first isolates the patient group with "High Cholesterol". Within that specific group, a significant majority of individuals "Had Heart Disease," further solidifying the link between high cholesterol and an increased risk of heart disease.

Shiny Dashboard Documentation

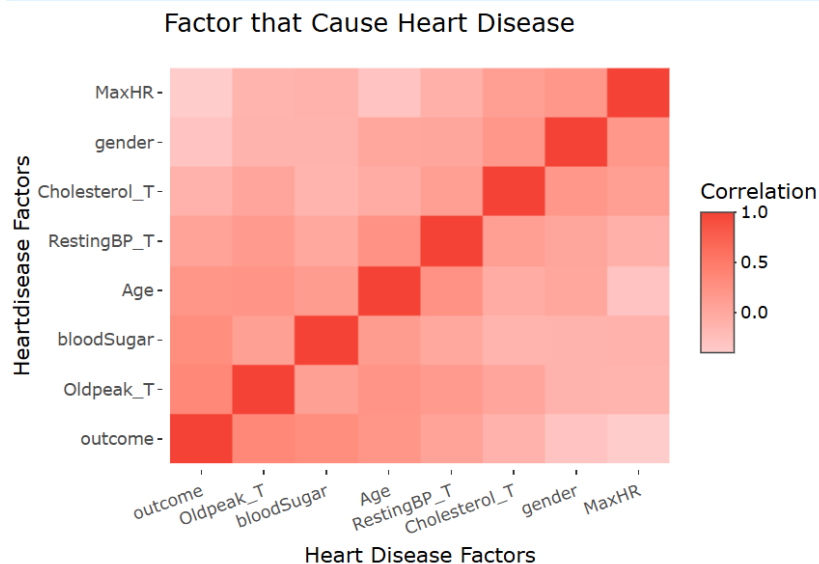
Link : https://kelompokaoi.shinyapps.io/DoesCholesterolCauseHeartDisease_AOL/

Our interactive Shiny dashboard, titled "Does Cholesterol cause Heart Disease?", is designed to provide a comprehensive analysis of heart disease and its contributing factors. The application is structured into three main pages to guide users from general information to specific data insights: Knowing Heart Disease, Patient Overview, and Conclusion.

1. Knowing Heart Disease

The initial section of the dashboard establishes the fundamental context for our research. The primary insight is that "Heart Disease" is not a single ailment but a broad term for a range of serious conditions affecting the heart's structure and function, including Coronary Artery Disease (CAD), Arrhythmias, and Heart Failure. The analysis then immediately highlights the severity of this issue, positioning it as the world's leading killer. Our findings show that heart and circulatory diseases caused an estimated 20 million deaths in 2021, which translates to nearly one in three deaths globally. This establishes the gravity of the problem we are analyzing.

This panel shows some factor of heart disease from our heatmap ;

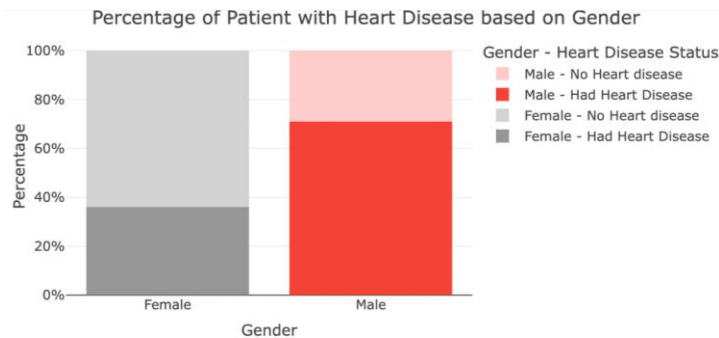


From here, Heart disease has 5 factors and that are blood pressure, blood sugar, age, cholesterol and gender.

Here are the explanation of gender and blood sugar plots.

1. Gender

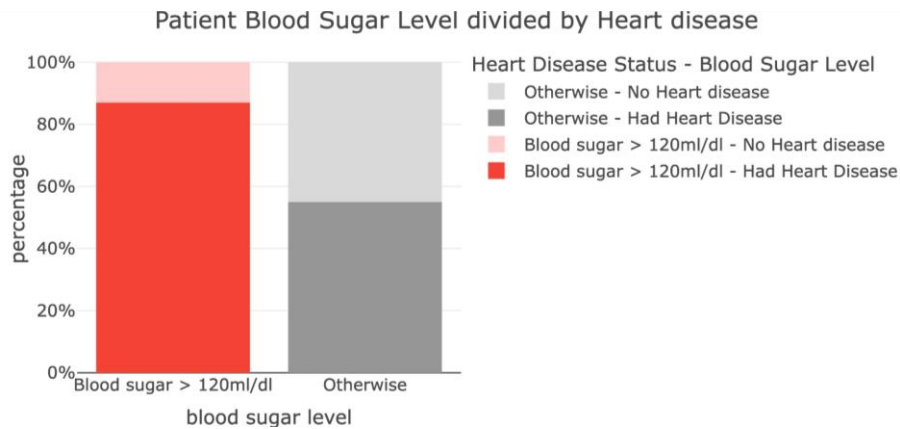
The bar chart below shows a comparison between individuals with heart disease status divided by gender. The bar chart shows a significant proportion difference between male individuals and female individuals with heart disease. Data shows that men have higher percentages of having heart disease.



- Gender Categories :
 - Male - Had heart disease
Individuals with the gender male and have had heart disease
 - Male - No heart disease
Individuals with the gender male and have never had heart disease
 - Female - Had heart disease
Individuals with the gender female and have had heart disease
 - Female - No heart disease
Individuals with the gender female and have never had heart disease
- Using a stacked bar chart gives a clear visualization of the proportion difference between female category and male category heart disease status:
 - X axis represents gender categories
 - Y axis shows the percentages of the sub-category (heart disease status) in the bar chart
- This bar chart shows the percentage of individuals with heart disease and no heart disease divided into gender groups.
 - Male
 - Had Heart Disease : 71.0%
 - No Heart disease : 29.0%
 - Female
 - Had Heart Disease : 36.0%
 - No Heart disease : 64.0%
- Insight :
The bar chart shows that men are more likely to suffer from heart disease than women with percentages of men with heart disease is almost double the percentage of women suffering the disease. Over the majority of men have heart disease while even less than 50% of women according to the data had heart disease.

2. Blood sugar levels

This data was extracted from the fasting blood sugar column, which marks individuals with or without blood sugar levels higher than 120 ml/dl. High blood sugar levels relate to the person's diabetic status. The bar chart below gives a comparison between heart disease status in individuals with blood sugar levels higher and otherwise.



- Blood sugar level Categories :
 - Blood sugar > 120ml/dl - Had heart disease
Blood sugar level is higher than 120ml/dl and have had heart disease
 - Blood sugar > 120ml/dl - No heart disease
Blood sugar level is higher than 120ml/dl and have never had heart disease
 - Otherwise - Had heart disease
Blood sugar levels not higher than 120ml/dl and have had heart disease
 - Otherwise - No heart disease
Blood sugar levels not higher than 120ml/dl and have never had heart disease
- The bar chart used shows clear proportion difference between individuals with or without heart disease divided by their blood sugar level:
 - X axis represents blood sugar level
 - Y axis shows the percentages comparing individuals with or without heart disease based on their blood sugar levels.
- This bar chart shows the rounded percentage of individuals with heart disease and no heart disease divided by blood sugar levels.
 - Blood sugar > 120ml/dl
 - Had Heart Disease : 87.0%
 - No Heart disease : 13.0%

- Otherwise
 - Had Heart Disease : 55.0%
 - No Heart disease : 45.0%
- Insight :

Based on the given data, people who have higher levels of blood sugar also have higher percentages of suffering from heart disease, compared to individuals who don't have blood sugar levels higher than 120ml/dl. This chart shows that blood sugar level may have an impact towards heart disease status.

2. Patient Overview

This tab gives users a thorough insight on the patients overview such as their cholesterol levels, the risk of getting heart diseases and their symptoms, including where the data came from. A summary of the results are written under each graph which provides users a better understanding of the visualisations given.

The beginning of the tab introduces the user to the source of data.

Where does the source from?

The data patient are collected from 5 large health institution. This large institution specialize in heart disease and health research.

List of institutions:

1. Hungarian Institute of Cardiology
2. University Hospital: Zurich
3. University Hospital: Basel
4. Medical Center, Long Beach
5. Cleveland Clinic Foundation

Then, the tab shows a user breakdown such as the amount of patients with high cholesterol and the percentage of patients with heart diseases. Users can choose to see how many patients are there per cholesterol level by toggling the Choose cholesterol level button.

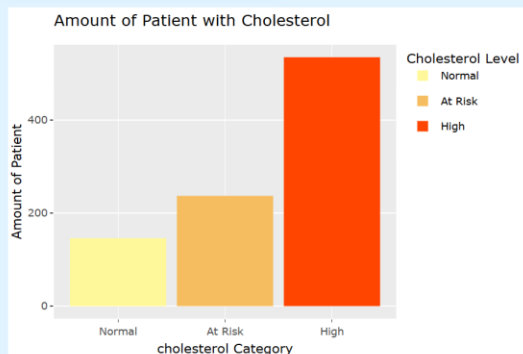
Breakdown of Patient Cholesterol Levels

Cholesterol is a known significant factor to heart diseases, which is one of the leading causes of deaths worldwide. High cholesterol increase the risk of getting a heart disease significantly. Cholesterol causes artery clogging which results in elevated levels of heart beats per minute or arrhythmias. Furthermore it leads to potential ruptures and internal bleeding behind the accumulated fat in the vessels.

Here is the distribution of Cholesterol between patient.

Choose Cholesterol Level :

☐ High ☐ Normal ☐ At Risk

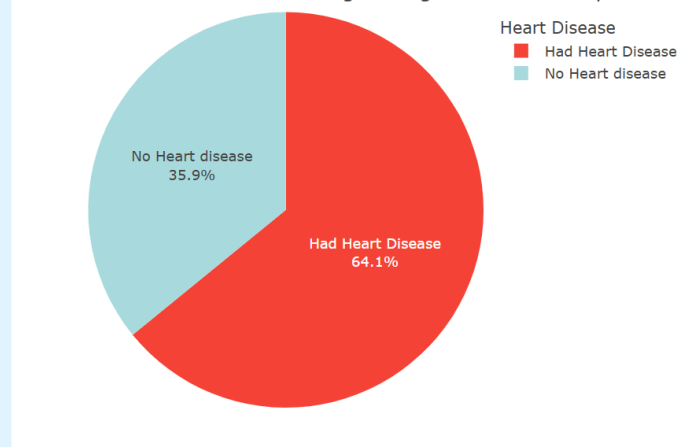


Many of the patient has high cholesterol

Heart disease in High Cholesterol Population

Could High cholesterol levels be increasing your risk of heart disease?

Patient Heart Disease Percentage in High Cholesterol Population



There are 64.1% of patient with high cholesterol has heart disease.

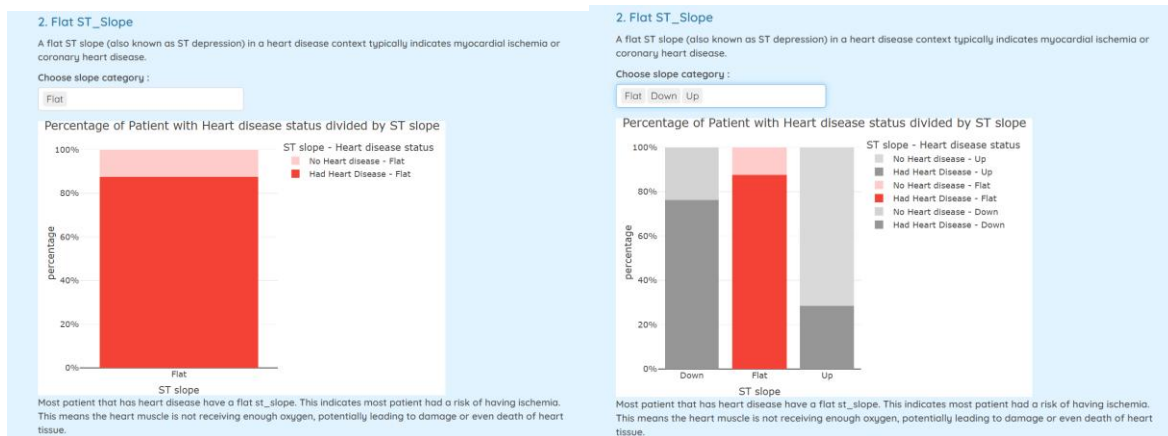
More than half of the patient had heart disease. High cholesterol does have an impact on heart disease. Increased cholesterol does cause heart disease. When cholesterol levels are too high, they can build up in the arteries, forming plaque that narrows and hardens the arteries, potentially leading to blocked blood flow and causing chest pain (angina) or a heart attack.

This tab also provides a symptom overview of patients that includes:

- Exercise angina
- ST slopes
- Maximum heart rates

- Oldpeak indicator
- Whether the patients are asymptomatic or not of chest pain

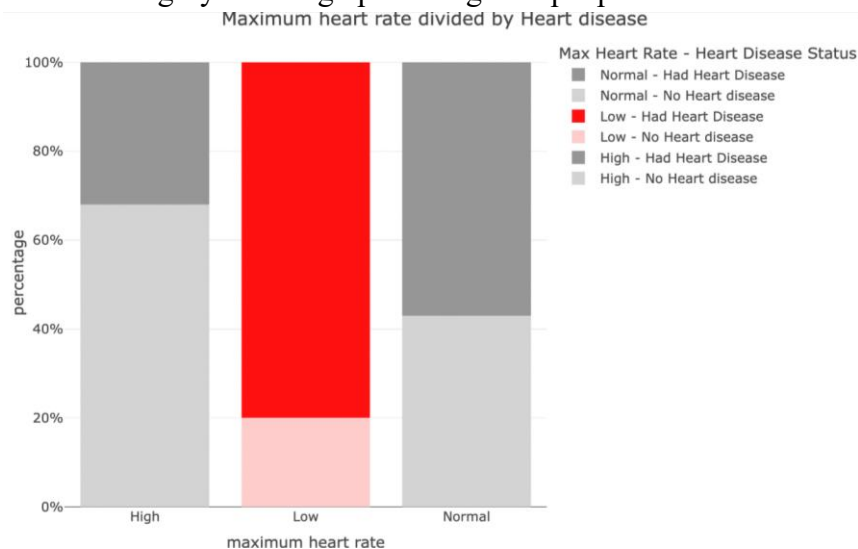
These graphs shows the emphasized graph in default, however toggling on the button gives user the ability to compare one graph and the other



Here are the explanation for each plot of symptoms

1. Maximum Heart Rate

This data was taken from an integer data documenting a person's maximum heart rate number. Maximum heart rate number is the maximum rate an individual's heart can achieve from exercising. This data is then separated into 3 categories based on its value and what the value indicates. Based on the graph below, it is known that the maximum heart rate in the low category have high percentages of people with heart disease.



● Maximum Heart Rate Categories :

- Low - Had heart disease

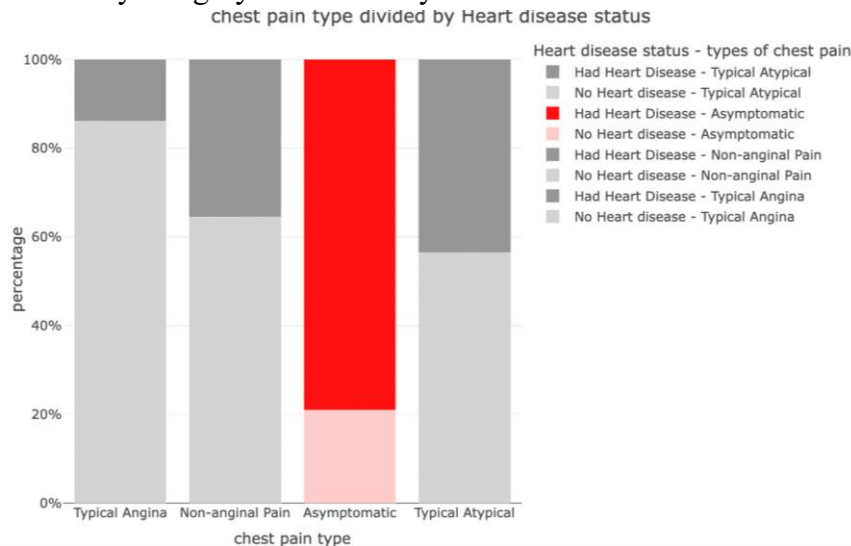
Maximum heart rate is lower than 60 and have had heart disease

- Low - No heart disease
Maximum heart rate is lower than 60 and have never had heart disease
 - Normal - Had heart disease
Maximum heart rate is higher or equal to 60 but less than 100 and have had heart disease
 - Normal - No heart disease
Maximum heart rate is higher or equal to 60 but less than 100 and have never had heart disease
 - High - Had heart disease
Maximum heart rate is higher than 100 and have had heart disease
 - High - No heart disease
Maximum heart rate is higher than 100 and have never had heart disease
- The bar chart is used to show different proportions of people with heart disease based on each category of their maximum heart rate:
 - X axis represents maximum heart rate category
 - Y axis shows the percentages comparing individuals with or without heart disease based on each maximum heart disease category.
 - Below is the percentages of people with or without hear disease based on their maximum heart rate
 - Low
 - Had Heart Disease : 80.0%
 - No Heart disease : 20.0%
 - Normal
 - Had Heart Disease : 57.0%
 - No Heart disease : 43.0%
 - High
 - Had Heart Disease : 32.0%
 - No Heart disease : 68.0%
 - Insight :

The proportions based on each bar on the graph shows that people with low maximum heart rate are mostly individuals with heart disease. Whereas people with maximum heart rate are less likely to be individuals with heart disease shown by only 32% of people from high maximum heart rate category having heart disease. Compared to 80% of people from low maximum heart rate category suffers from heart disease.

2. Chest Pain

Out of all chest pain categories below, Asymptomatic chest pain is known to be caused by heart disease. And the bar chart below shows that people with asymptomatic chest pain are mostly people with heart disease. In comparison to other chest pain types, asymptomatic is the only category dominated by individuals with heart disease.



- Chest pain type Categories :
 - Typical Angina
Chest pain gained from physical exertion
 - Non-Anginal pain
Chest pain gained not related to blood flow issues
 - Asymptomatic
Chest pain gained without symptoms
 - Typical Atypical
Chest pain with angina-like characteristics
- The bar chart shows the types of chest pain people may experience divided by heart disease status:
 - X axis represents chest pain category
 - The Y axis shows the percentages of people based on their heart disease status.
- Below is the percentages of people with or without heart disease based on their maximum heart rate
 - Typical Angina
 - Had Heart Disease : 13.87%
 - No Heart disease : 86.12%
 - Non-Anginal pain

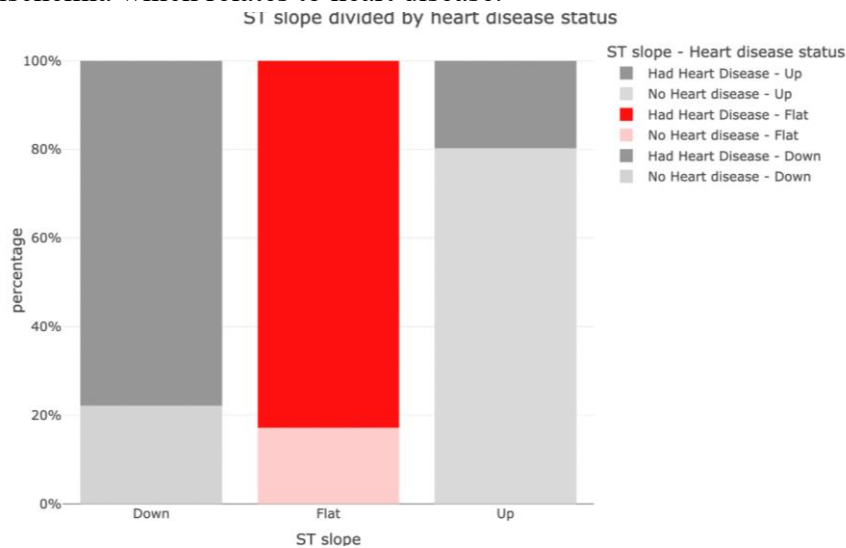
- Had Heart Disease : 35.46%
- No Heart disease : 64.53%
- Asymptomatic
 - Had Heart Disease : 79.03%
 - No Heart disease : 20.96%
- Typical Atypical
 - Had Heart Disease : 43.47%
 - No Heart disease : 56.52%

- Insight :

The chart above gives a clear visual of which type of chest pain is majorly experienced by people with heart disease. It is shown that the majority of people suffering from heart disease experience asymptomatic chest pain. Asymptomatic chest pain is a type of chest pain experienced without symptoms and can be a key indicator towards heart disease.

3. ST slope

Below is a bar chart showing ST slope data received from people with and without heart disease. The bar chart shows the proportion difference between flat slope and down slope category compared to up slope category. Down or flat ST slope can be signs of ischemia which relates to heart disease.



- ST slope Categories :

- Down - Had heart disease
ST segment descends and have had heart disease
- Flat - Had heart disease
ST segment is flat and have had heart disease
- Flat - No heart disease

- ST segment is flat and have never had heart disease
 - Up - Had heart disease
ST segment rises and have had heart disease
 - Up - No heart disease
ST segment rises and have never had heart disease
- The bar chart shows the population of people with heart disease in each category of ST slope:
 - X axis represents ST slope category
 - The Y axis shows the percentages of people based on their heart disease status.
- Below is the percentages of people with or without heart disease based on the ST slope segment
 - Down
 - Had Heart Disease : 77.77%
 - No Heart disease : 22.22%
 - Flat
 - Had Heart Disease : 82.82%
 - No Heart disease : 17.17%
 - Up
 - Had Heart Disease : 19.74%
 - No Heart disease : 80.25%
- Insight :

Based on the chart above, it can be concluded that people with heart disease have a very small likelihood of having an up slope ST segment. Down slope can be a strong indicator of ischemia, which is reduced blood flow, and is a sign of heart disease. Down slope and flat slope categories are dominated by people with heart disease, with about 77% of the population from the down slope category and 82% from the flat slope category having heart disease.

4. Exercise Angina

This chart shows how many people from each category (having heart disease and not having heart disease) experience exercise angina. The bar chart shows the percentages of people from each category and divides the bar based on whether they experience exercise angina or don't experience exercise angina.



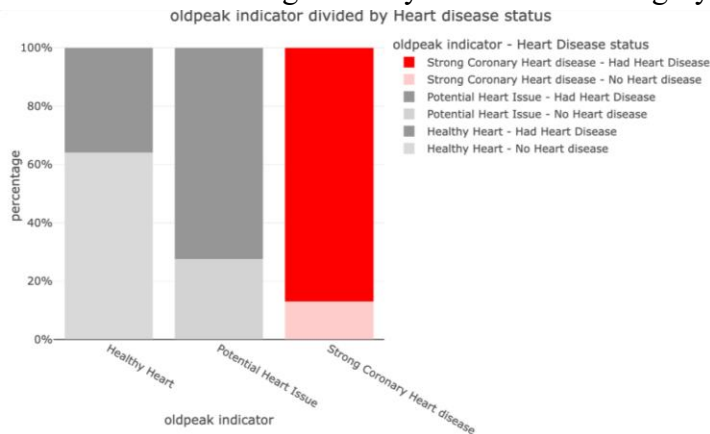
- Categories :
 - Yes - Had heart disease
Experiences exercise angina and have had heart disease
 - Yes - No heart disease
Experiences exercise angina and have never had heart disease
 - No - Had heart disease
Does not experience exercise angina and have had heart disease
 - No - No heart disease
Does not experience exercise angina and have never had heart disease
- The bar chart shows the percentages from the whole population in each category and divided the category based on whether they experience exercise angina:
 - X axis represents Heart disease status
 - The Y axis shows percentages of the whole population divided by heart disease category.
- Below is the percentages of people with or without hear disease based on their maximum heart rate
 - Had Heart Disease
 - Yes : 34.0%
 - No : 21.0%
 - No Heart Disease
 - Yes : 6.0%
 - No : 39.0%

- Insight :

The chart shows the majority of the whole population (55%) has heart disease, whereas the rest 45 does not. And the majority of the people in the heart disease category experiences exercise angina. And only a tiny percentage of people for about only 6% experiences exercise angina without having heart disease. By this information we can assume that people who experience exercise angina are more likely to be having heart disease.

5. Oldpeak

The bar chart shows the distribution of people with heart disease based on the oldpeak category. The oldpeak data was taken from numeric data and is converted into factors based on their value category. The chart below shows that the majority of people with heart disease lie in the strong coronary heart disease category.



- Oldpeak Categories :

- Healthy Heart - Had heart disease
Oldpeak value equal to 0.0 and have had heart disease
- Healthy Heart - No heart disease
Oldpeak value equal to 0.0 and have had heart disease
- Potential Heart Disease - Had heart disease
Oldpeak value higher than 0.0 but lower than 2.0 and have had heart disease
- Potential Heart Disease - No heart disease
Oldpeak value higher than 0.0 but lower than 2.0 and have never had heart disease
- Strong Coronary Heart Disease - Had heart disease
Oldpeak value higher than or equal to 2.0 and have had heart disease
- Strong Coronary Heart Disease - No heart disease
Oldpeak value higher than or equal to 2.0 and have never had heart disease

- The bar chart is used to show different proportions of people with heart disease based on each category of their oldpeak value:
 - X axis represents oldpeak category
 - The Y axis shows the percentages of people divided by heart disease status.
- Below is the percentages of people with or without heart disease based on their oldpeak category
 - Healthy Heart
 - Had Heart Disease : 35.87%
 - No Heart disease : 64.12%
 - Potential Heart Disease
 - Had Heart Disease : 72.42%
 - No Heart disease : 27.57%
 - Strong Coronary Heart Disease
 - Had Heart Disease : 86.93%
 - No Heart disease : 13.06%
- Insight :

Based on the bar chart above, it is shown that the category mostly dominated by people with heart disease is the strong coronary heart disease category with over 86%. While the healthy heart category only has 35% of people suffering from heart disease. This data shows that oldpeak value may be a strong indicator of heart disease.

3. Conclusion

The last tab provides users with a brief conclusion of the SDG Goals supporting the research, the research findings and a few ways to prevent it.

Is Cholesterol The Culprit Behind Heart Disease?	Knowing Heart Disease	Patient Overview	Conclusion
The Sustainable Development Goal, adopted by all United Nations Member States in 2015, provides a shared blueprint for peace and prosperity for people and the planet, now and into the future. At its heart are the 17 Sustainable Development Goals (SDGs), which are an urgent call for action by all countries - developed and developing - in a global partnership. Based on Goal 3, Good Health and well being are adopted to ensure healthy lives and promote well being for all ages. One of its sub-target is Goal 3.4.1 Mortality rate attributed to cardiovascular disease, cancer, diabetes or chronic disease in hope to reduce premature mortality from those diseases by 2030. Heart disease are the leading cause of Non communicable disease related premature death globally. Since SDG 3.4.1 targets reduction of deaths from NCDs, addressing heart disease is a critical component. Heart disease preventive efforts are encouraged by sdg 3.4.1 From the patient data, we can conclude that cholesterol does cause heart disease. People with high cholesterol have an increased the risk of having a heart disease. Cholesterol is an important factor that cause heart disease. But it's not enough to look at cholesterol alone. Heart disease is complex. There are other factor of heart disease like, age, blood pressure, blood sugar, and more that shouldn't overlook. Apart from the risk factor, it is also important to identify the symptoms of heart disease that an individual have. By detecting and treating heart conditions early this can reduce the risk of premature death Start living a well-rounded healthy life—eat a balanced diet, stay active, get enough sleep, and have regular health check-ups. Let's stay informed and stay healthy.			